Applied Acoustics 178 (2021) 107969

Contents lists available at ScienceDirect

Applied Acoustics

journal homepage: www.elsevier.com/locate/apacoust

Teaching-learning-based optimization of an ultra-broadband parallel sound absorber

Nansha Gao [a,b,*], Baozhu Wang [c], Kuan Lu [d], Hong Hou [a]

[a] Key Laboratory of Ocean Acoustic and Sensing, School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China
[b] Division of Physics and Applied Physics, School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore 637371, Singapore
[c] Qingdao Airjj New Materials Co. Ltd., Qingdao 266100, China
[d] School of Mechatronics Engineering, North University of China, Taiyuan 030051, China

ARTICLE INFO

Article history:
Received 30 November 2020
Received in revised form 27 January 2021
Accepted 1 February 2021
Available online 26 February 2021

Keywords:
Broadband sound absorption
TLBO algorithm
Effective medium model
Transfer matrix method

ABSTRACT

The average sound-absorption coefficient of an ultra-broadband parallel sound absorber, between 0 and 2000 Hz, is optimized in certain conditions. This is accomplished using a teaching–learning-based optimization algorithm, where the geometric parameters are used as optimization variables. The optimized ultra-broadband parallel sound absorber has an average sound-absorption coefficient of 0.9255. The surface impedance of its structure is almost perfectly matched with the surrounding air – thanks to the optimized lengths of both the lateral and micro-perforated plates. The effective-medium model, the transfer-matrix method, and the finite-element method are applied and show consistent results. The acoustic-impedance tube measurements confirm that the optimized ultra-broadband parallel sound absorber shows high sound-absorption from 200 to 1715 Hz. The investigated passive acoustic design opens new possibilities for new absorber types in the future.

1. Introduction

Effective absorption of medium- and low-frequency broadband sound-waves has been an important research topic for many years [1]. It is particularly important for architectural acoustics as well as vibration- and noise-reduction [2,3]. Foam [2,3], fiber [4–6], a sandwich structure [7], and 3D-printed porous structures [8], which have holes inside, can effectively dissipate high-frequency sound waves into heat energy. A common classical structure, micro-perforated plates [9], can greatly improve the sound-absorption peak of both low and medium frequencies. However, their performance is poor for high frequencies. Membrane-type acoustic metamaterials [10,11] enable broadband sound-absorption by utilizing the strain-energy at internal boundaries. Unfortunately, the preload force cannot ensure consistency during the assembly process, and the membrane is prone to both fatigue and aging. Multi-cavity resonance [12–16] can also be used to achieve broadband and quasi-perfect sound-absorption. The effect of optimization [17–19] when designing an acoustic structure is to shorten the product development cycle and improve their sound absorption performance. Compared to a genetic- or particle-swarm-optimization algorithm [20,21], a teaching–learning based (TLBO) optimization algorithm [22] has fewer control-parameters, a more accurate solution, and it converges better. In our earlier studies [23–25], theoretical calculation and experimental results show that the sound absorption peak can be improved and widened by the combination of porous material and lateral plate, but their low-frequency sound absorption performance is poor.

In this study, we designed and built an ultra-broadband parallel sound-absorber (UBPSA) that uses a combination of porous structures, lateral plates, and micro-perforated panels. By selecting the average sound-absorption coefficient as the target, the size of the structure was optimized with the TLBO algorithm. The results of the theoretical model show that the optimized UBPSA could achieve an average sound-absorption coefficient of 0.9255 from 200 to 1750 Hz. These results were verified using the finite element method (FEM) and an acoustic-impedance tube test.

2. Structure and methods

2.1. UBPSA design

The UBPSA proposed in this work consists of sound-absorbing bodies on the left and right, which are aligned parallel – see Fig. 1(b). The left part consists of a micro-perforated plate with a

* Corresponding author at: Key Laboratory of Ocean Acoustic and Sensing, School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China.

E-mail address: gaonansha@nwpu.edu.cn (N. Gao).

https://doi.org/10.1016/j.apacoust.2021.107969
0003-682X/

cavity and a porous four-layer composite material with the thickness $H2$. The right part consists of a porous four-layer composite material with the thickness $H5$. The thicknesses of the porous composite structures are $c1$-$c8$, while the lengths of the lateral plates are $b1$-$b8$, and their thickness is t. The thickness of the micro-perforated plate is $H4$, the aperture is a, the hole spacing is m, and the depth of the rear cavity is $H3$. The widths of the left and right parts are $W1$ and $W2$, respectively. The porous materials in the left and right halves are different (material 1 and 2). The specific dimensions are shown in Table 1.

2.2. Effective medium model

To describe the propagation and dissipation of sound waves in rigid frameworks, which are made of porous materials, the porous materials can be regarded as equivalent fluid, by using a semi-empirical parametric model known as the Johnson-Champoux-Allard (JCA) [26] model. The five acoustic parameters, which are used in the JCA model, are: flow resistivity (σ), porosity (φ), tortuosity (α_∞), viscous characteristic length (Λ) and thermal characteristic length (Λ'). Fig. 2a and b show the samples of porous materials 1 and 2. Industrial computerized tomography (CT) images display the honeycomb like cavities with uniform size distribution with the aid of micro-nano tomographic imaging system (Phoenix Vtomex-S, Shanghai Yinghua Inspection and Testing Co., Ltd). The five acoustic parameters (see Table 2) are provided by Baochi Phonics Technology Co., Ltd.

Effective density $\rho_{p,e}$ and modulus $K_{p,e}$, which vary with frequency, can be determined using the above acoustic parameters. Their relationships are [26]:

$$\rho_{p,e} = \frac{\rho_0\alpha_\infty}{\phi}\left(1 + i\frac{\omega_v}{\omega}F(\omega)\right) \tag{1}$$

$$K_{p,e}^{-1} = \frac{\gamma P_0}{\phi(\gamma - (\gamma-1)(1 + i(\omega_c'/(\mathrm{Pr}\omega))G(\mathrm{Pr}\omega))^{-1})} \tag{2}$$

$$\mathrm{F}(\omega) = \sqrt{1 - i\eta\rho_0\mathrm{Pr}\omega\left(\frac{2\alpha_\infty}{\sigma\varphi\mathrm{A}}\right)^2} \tag{3}$$

$$\mathrm{G}(\mathrm{Pr}\omega) = \sqrt{1 - i\eta\rho_0\mathrm{Pr}\omega\left(\frac{2\alpha_\infty}{\sigma'\phi\Lambda'}\right)^2} \tag{4}$$

Here, the density of air is ρ_0 = 1.213 kg/m^3, Pr = 0.71 is the Prandtl number, P_0 is the atmospheric pressure, the specific-heat ratio of air is γ = 1.4, and the dynamic viscosity of air is η = 17.9 $\times$ 10^{-6} Pa•s. The Biot frequency, ω_v, adiabatic cross-over angular frequency, ω_c', and the thermal resistivity σ' used in Eqs. (1) and (2) are defined as follows [26]:

$$\omega_v = \sigma\phi/\rho_0\alpha_\infty \tag{5}$$

$$\omega_c' = \sigma'\phi/\rho_0\alpha_\infty \tag{6}$$

$$\sigma' = 8\alpha_\infty\eta/\phi\Lambda' \tag{7}$$

Since the lateral plates are distributed within the porous materials, the effective density $\rho_{e,i}$ and modulus $K_{e,i}$ can be described as follows [25,27,28]:

$$\rho_{e,i,x} = \infty \tag{8-1}$$

$$\rho_{e,i,y} = \rho_{p,e}/\eta_i \ (i = 1, 2, ...8) \tag{8-2}$$

$$K_{e,i} = K_{p,e}/\eta_i \ (i = 1, 2, ...8) \tag{8-3}$$

Here, the parameter η_i is the volume ratio for the lateral plate to porous material in the effective layers 1–8 – see Fig. 2(b). The effective impedance of the effective layers is defined using the following expressions [25,29]:

$$c_{e,i} = \sqrt{K_{e,i}/\rho_{e,i,y}} \ (i = 1, 2, ...8) \tag{9-1}$$

$$z_{e,i} = \rho_{e,i,y} \cdot c_{e,i} \ (i = 1, 2, ...8) \tag{9-2}$$

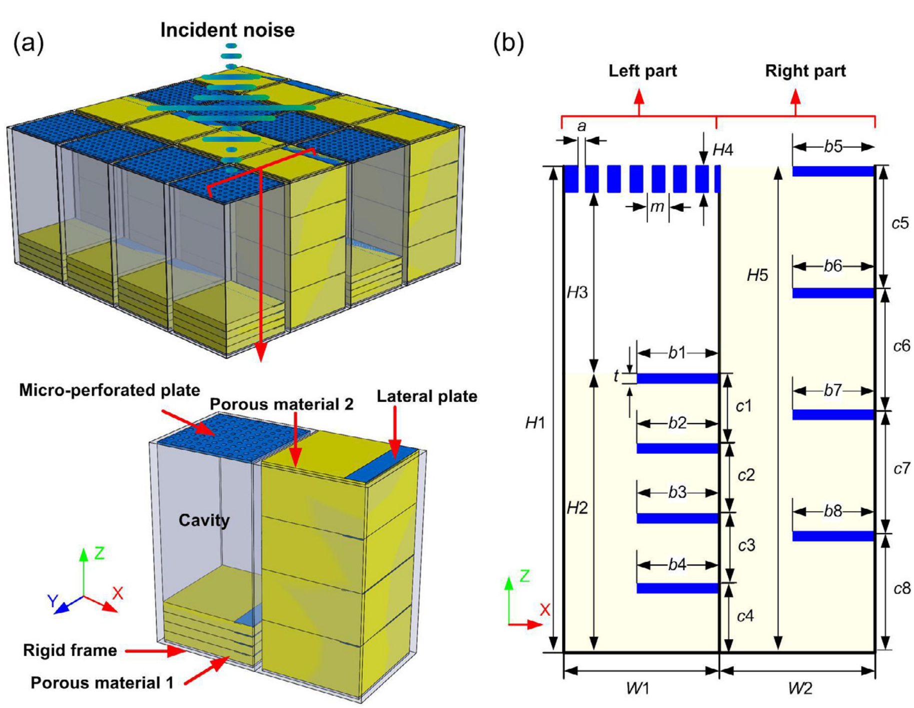

Fig. 1. (a) Schematic of the UBPSA. (b) Geometric dimensions on XZ plane.

Table 1
Geometrical dimensions for the non-optimized UBPSA. (all values in mm).

	*H*1	*H*2	*H*3	*H*4	*H*5
Non-optimized	126	24	100	2	60
	*b*1	*b*2	*b*3	*b*4	*b*5
Non-optimized	25	25	25	25	20
	*b*6	*b*7	*b*8	*c*1	*c*2
Non-optimized	20	20	20	6	6
	*c*3	*c*4	*c*5	*c*6	*c*7
Non-optimized	6	6	15	15	15
	*c*8	*a*	*m*	*W*1	*W*2
Non-optimized	15	0.6	5	48	48

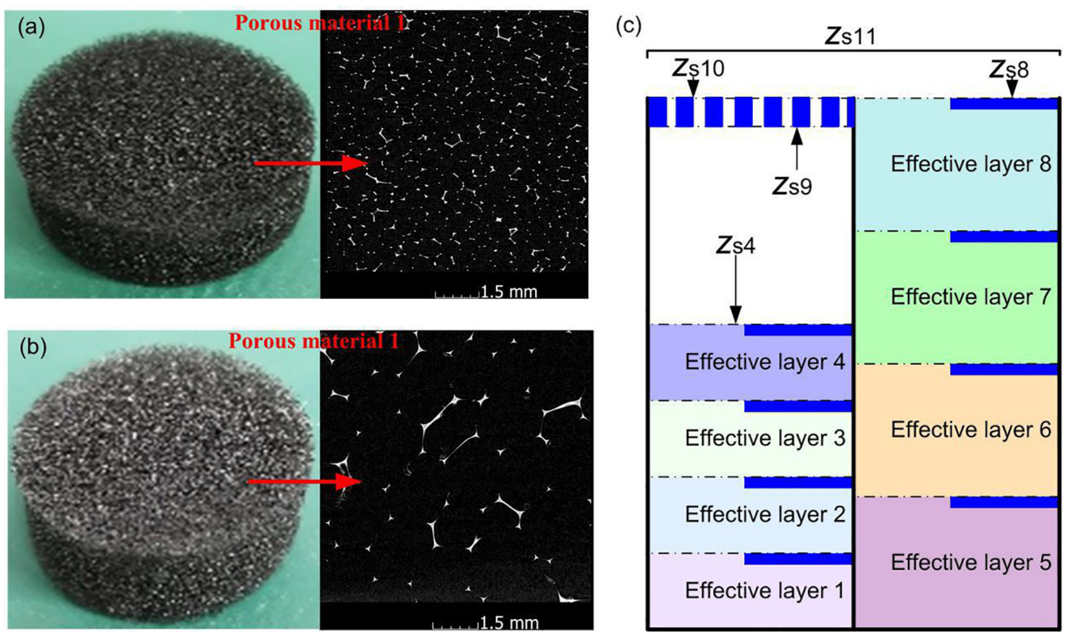

Fig. 2. (a) and (b) Industrial CT images of the used porous materials, (c) Effective medium model for the UBPSA.

Table 2
Acoustic parameters used in the JCA model for the two porous materials.

	φ	σ [Pa·s·m^{-3}]	α_∞	Λ [m]	Λ' [m]
Porous material 1	0.98	6000	1.02	591×10^{-6}	323×10^{-6}
Porous material 2	0.983	2100	1.08	532×10^{-6}	189×10^{-6}

2.3. TM method

In case of normal incidence of the plane wave, both sound pressure and particle velocity are assumed continuous. Therefore, the following matrix can be used to describe the relationship between sound pressure and particle velocity in each layer of the effective medium [29]:

$$\begin{bmatrix} p_{t,i} \\ u_{t,i} \end{bmatrix} = \begin{bmatrix} \cos(k_{e,i,y} \cdot c_i) & i\frac{\omega\rho_{e,i}}{k_{e,i,y}}\sin(k_{e,i,y} \cdot c_i) \\ i\frac{k_{e,i,x}}{\omega\rho_{e,i}}\sin(k_{e,i,y} \cdot c_i) & \cos(k_{e,i,y} \cdot c_i) \end{bmatrix} \begin{bmatrix} p_{b,i} \\ u_{b,i} \end{bmatrix} \tag{10}$$

Here, ($p_{t,i}$, $u_{t,i}$) and ($p_{b,i}$, $u_{b,i}$) correspond to the sound pressures and particle velocities of the incident and exit planes in the i-th layer of the effective medium, respectively. The thickness of the effective medium is c_i. The effective wave number in each layer of the UBPSA is [29]:

$$k_{e,i,y} = \omega/\sqrt{K_{e,i}/\rho_{e,i,y}} \tag{11}$$

The bottom of the UBPSA ensures total reflection of the incident sound wave. Hence, the impedances of the effective layer 1,5 and porous material are

$$z_{e,i} = -z_{p,e} \cdot \cot(k_{e,i} \cdot c_i)\ (i = 1, 5) \tag{12-1}$$

$$z_{p,e} = \sqrt{K_{p,e} \cdot \rho_{p,e}} \tag{12-2}$$

As shown in Fig. 2(b), the surface impedances $z_{s,4}$, and $z_{s,8}$ are [30]:

$$z_{s,4} = \frac{-iz_{s,3}z_{e,4}\frac{k_{p,e}}{k_{e,i,y}}\cot(k_{e,i,y(i=1,2,3,4)} \cdot ci) + \left(z_{p,e} \cdot \frac{k_{p,e}}{k_{e,i,y}}\right)^2}{z_{s,3} - iz_{e,4}\frac{k_{p,e}}{k_{e,i,y}}\cot(k_{e,i,y(i=1,2,3,4)} \cdot ci)}(i = 1, 2, 3, 4) \tag{13-1}$$

$$z_{s,8} = \frac{-iz_{s,7}z_{e,8}\frac{k_{p,e}}{k_{e,i,y}}\cot(k_{e,i,y} \cdot ci) + \left(z_{p,e} \cdot \frac{k_{p,e}}{k_{e,i,y}}\right)^2}{z_{s,7} - iz_{e,8}\frac{k_{p,e}}{k_{e,i,y}}\cot(k_{e,i,y} \cdot ci)}(i = 5, 6, 7, 8) \tag{13-2}$$

The effective wave number $k_{p,e}$ in the porous material is [30]:

$$k_{p,e} = \frac{\omega}{\sqrt{K_{p,e}/\rho_{p,e}}} \tag{14}$$

The surface impedance z_{s9} for the lower end of a micro-perforated plate can be determined using the following transfer-matrix [30]:

$$z_{s9} = \frac{-z_{s4} j \rho_0 c_0 \cot(k_0 \cdot H_3) + z_0^2}{z_{s4} - j \rho_0 c_0 \cot(k_0 \cdot H_3)} \tag{15}$$

Here, c_0 = 343 m/s is the speed of sound in air, $k_0 = \omega/c_0$ is the wave number in air, z_0 = 442.47 Pa.s/m^3. A micro-perforated plate with a cavity at the back can be treated like multiple Helmholtz secondary resonators in parallel to their necks. If loss occurs for a sound wave in the neck, the resistance r_m and additional mass m of the Helmholtz resonators can be calculated as follows [31]:

$$m = \frac{\rho}{\varepsilon}[H_4 + \delta \cdot a + \sqrt{\frac{8v}{\omega}}(1 + \frac{H_4}{a})] \tag{16}$$

$$r_m = \frac{\rho}{\varepsilon}\sqrt{8v\omega}\left(1 + \frac{H_4}{a}\right) \tag{17}$$

Here, $v = 15 \times 10^{-6}$ m^{2}s^{-1} is the kinemetric viscosity of air. δ, is the end correction factor, which, to a first approximation, is usually taken as 0.85 and derived by considering the radiation impedance of a neck. The micro-perforated plate in this study is a square plate with circular holes. Hence, δ is [30]:

$$\delta = 0.85(1 - 1.14\sqrt{\varepsilon}) \tag{18}$$

The Eqs. (16) and (17) can be used to calculate the surface impedance z_{s11} for the upper end of the micro-perforated plate [30]:

$$z_{s10} = r_m + j[\omega m - \rho_0 c_0 \cot(k_0 \cdot H_3)] \tag{19}$$

Since the left and right halves of the UBPSA are in parallel, z_{s11} can be written as:

$$z_{s11} = \frac{z_{s10} \cdot z_{s8}}{z_{s10} + z_{s8}} \tag{20}$$

The reflection coefficient ($R(\omega)$) and absorption coefficient ($\alpha(\omega)$) can be calculated using the following expressions [29–31]:

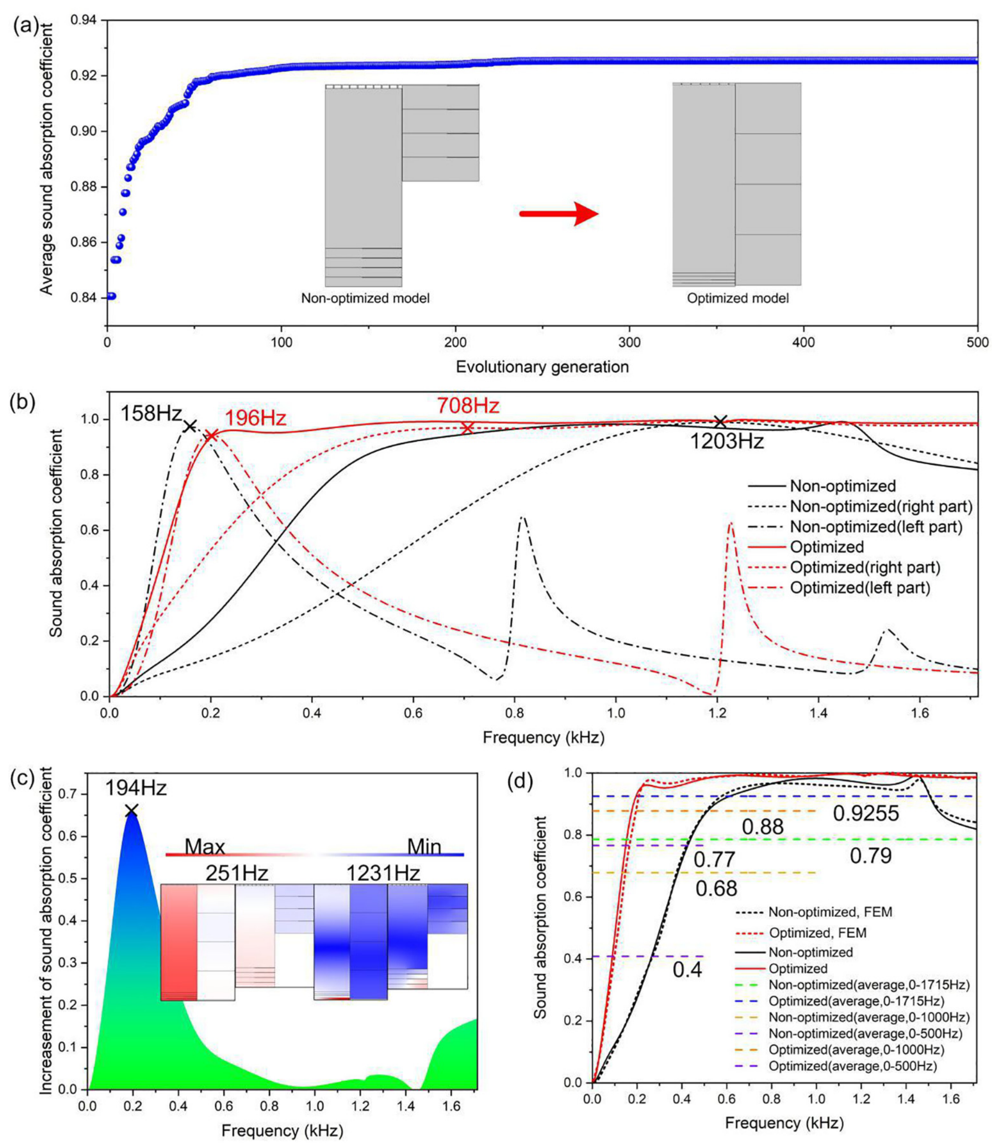

Fig. 3. (a) TLBO-algorithm optimization process. (b)–(d) Comparison of sound-absorption coefficients between optimized and non-optimized UBPSA.

$$R(\omega) = \frac{z_{s11}(\omega) - z_0}{z_{s11}(\omega) + z_0} \tag{21}$$

$$\alpha(\omega) = 1 - |R(\omega)|^2 \tag{22}$$

3. Results

3.1. Comparison of sound absorption

The TLBO algorithm [22] simulates the process of teachers imparting knowledge to students. The relationship between evolutionary generation in the TLBO algorithm and the average sound-absorption coefficient is shown in Fig. 3(a). One can see that the average sound-absorption coefficient becomes larger as the algorithm processes through successive generations and eventually converges to a (near) constant value. Fig. 4 shows that after approximately 100 generations of the TLBO algorithm, the ordinate on the curve converges to the optimal solution (0.9255). The optimized structural parameters are shown in Table 3.

Compared with the UBPSA, which was optimized by the TLBO algorithm, the thicknesses of both the effective layers 1–4 and the micro-perforated plate are thinner, which means the volume of the cavity increased. The effective layers 5–8 became thicker, and the insert plate disappeared. However, it should be noted that the total thickness of the UBPSA only increased from 126 to 151 mm. It can be seen from Fig. 3b that, without optimization, the sound-absorption curve of the left part reflects the characteristics of a typical micro-perforated plate. Its first broadband sound-absorption peak is at 158 Hz. The sound-absorption curve of the right part also conforms to the characteristics of porous materials, reaching a sound-absorption peak at about 1203 Hz. The reason the sound-absorption effect is not obvious in the middle and low frequency range after the left and right parts were connected in parallel is that the corresponding optimal sound-absorption frequency is far away. After TLBO optimization, the absorption peaks of the left and right parts of the UBPSA were 196 Hz and 708 Hz respectively. The overall sound-absorption curve is shown in Fig. 3b. Fig. 3c shows the increase of the sound-absorption coefficient, before and after TLBO optimization, within 1750 Hz. The increase at 194 Hz is more than 0.65, and the main increase values are distributed within 500 Hz. To check the correctness of the analytical model, we used the acoustic module of COMSOL Multiphysics® Modeling software [32,35] as the FEM to calculate the sound-absorption performance of the UBPSA – see the dotted line in Fig. 3d. For 0 to 1750 Hz, the calculation results of FEM and analytical model agree very well, which suggests the validity of the method. Fig. 3d also indicates that the average sound-absorption coefficient was improved from 0.4 to 0.77, below 500 Hz. Below 1000 Hz it improved from 0.68 to 0.88. In other words, the improvement of the average sound-absorption coefficient, when the TLBO optimization algorithm was used, occurred mainly for the middle and low frequencies. The sound pressure distribution of the UBPSA at 251 Hz and 1231 Hz, before and after TLBO optimization, is also shown in Fig. 3c. At low frequencies, most of the sound energy could still be concentrated in the cavity of the micro-perforated plate, while the porous material has no obvious effect on the absorption. At high frequencies, the absorption of sound waves by the micro-perforated plates can almost be omitted. The porous materials, however, play an important role. The high sound-pressure indicates that TLBO optimization can improve sound absorption.

3.2. Complex plane analysis and surface impedance

Fig. 4a reveals five zero points (marked as O1-O5 in order) near the real frequency axis of the reflection-coefficient logarithm. This

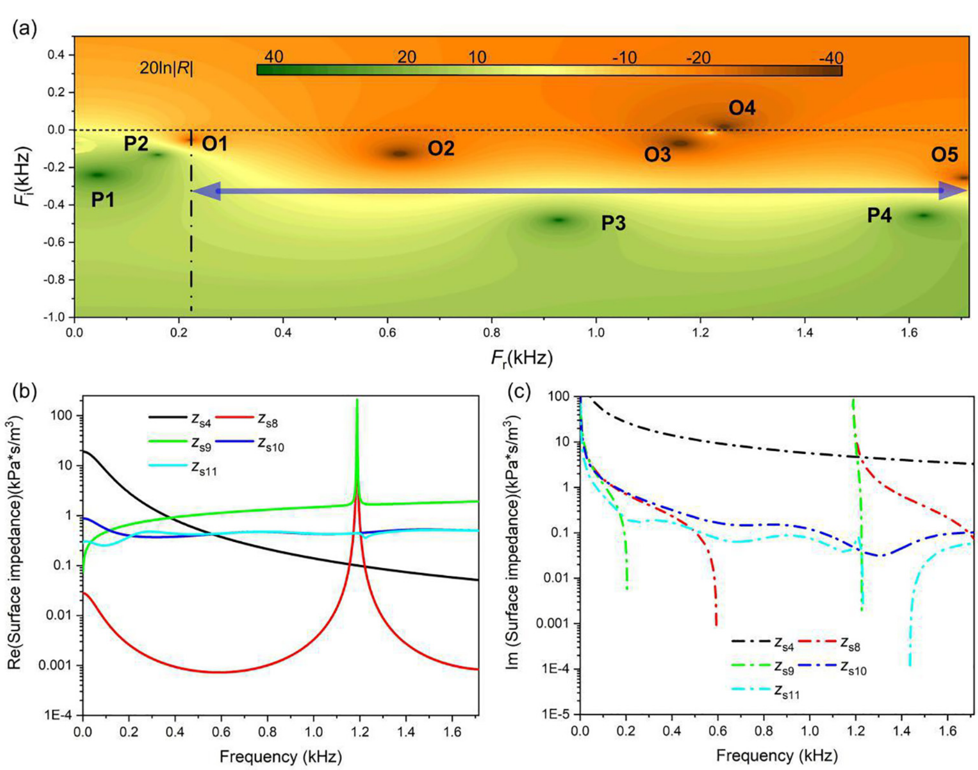

Fig. 4. (a) Distribution of sound pressure and particle velocity at points A and B in Fig. 3d. (b) Distribution of corresponding zero and pole points in the complex plane. (c) and (d) Real and imaginary parts of the effective surface impedances z_{s4}, z_{s8}, z_{s9}, $z_{s10,}$ and z_{s11}, respectively.

suggests the presence of sound-absorption peaks at these five frequencies. The zero points O1, O2, and O5 are located near the real frequency axis. The three frequency-points meet the requirements $Q_{leak} \approx Q_{loss}$ and can achieve quasi-perfect sound-absorption. Zero point O1 is close to the real frequency axis but its color is lighter, and its reflection coefficient is relatively large. This indicates that, although quasi-perfect sound-absorption can be achieved here, its corresponding sound-absorption coefficient is not very high. Compared with zero point O1, even though zero point O5 is relatively far away from the real frequency axis, its color is dark, the corresponding reflection-coefficient is low. As a result, the sound-absorption peak is higher than for zero point O1. Furthermore, the position is not only close to the real frequency axis but also its reflection coefficient is small. Hence, the sound-absorption coefficient is the highest among the three zero points (O1, O2, O5). The location of zero point O2 is not only close to the real frequency axis but also its reflection coefficient is low. Therefore, the sound-absorption coefficient is the highest among the three zero points (O1, O2, O5). The zero points O3 and O4 are very close to the real frequency axis, which means it can be assumed that perfect sound-absorption can be achieved here ($Q_{leak} = Q_{loss}$). Based on the location of the pole points, considering the internal loss of the UBPSA and the coupling effect between the pole points, the locations of the pole points differ from the position of zero points. In addition, there are four pole points in the operating-frequency range (marked P1-P4 in order), The imaginary part of the amplitudes (i.e. the distance from the pole points to the real axis) of frequencies of pole points P1 and P2 are lower. This indicates that the corresponding broadband sound-absorption effect is not near pole points 1 and 2. Instead the imaginary part of the amplitudes for pole points 3 and 4 are large, which indicates that the corresponding broadband sound-absorption effect is better at the frequencies near pole points 3 and 4. The analysis results of zero and pole points on complex frequency plane suggest that the prediction for the sound-absorption coefficient of the UBPSA is consistent with the theoretical calculation results. The results in Fig. 4b and c show that the real and imaginary parts of the surface impedances z_{s4}–z_{s11} vary with frequency. As can be seen in Fig. 4b, the real parts of z_{s10} and z_{s11} are very close to the air impedance for the whole frequency-range. The real part of z_{s8} and z_{s9} has a peak at 1189 Hz, and their imaginary parts change from negative to positive at this frequency. Therefore, they are associated with quasi anti-resonance absorption [33,36]. Together with Fig. 4a, it can be concluded that the reason for the perfect sound-absorption is quasi anti-resonance. The acoustic surface impedance z_{s11} determines the change of the reflection coefficient $R(\omega)$. Owing to the combination of various parts of the optimized UBPSA, the imaginary part of z_{s11} (acoustic reactance) is 0, which perfectly matches the impedance of air.

Table 3
Geometrical dimensions of the optimized UBPSA. (all values in mm).

	*H*1	*H*2	*H*3	*H*4	*H*5
Optimized	151	10	140	1	150
	*b*1	*b*2	*b*3	*b*4	*b*5
Optimized	0.71	0	18.11	39.9	0
	*b*6	*b*7	*b*8	*c*1	*c*2
Optimized	0	0	0	2.5	2.5
	*c*3	*c*4	*c*5	*c*6	*c*7
Optimized	2.5	2.5	37.5	37.5	37.5
	*c*8	*a*	*m*	*W*1	*W*2
Optimized	37.5	0.32	7	47	49

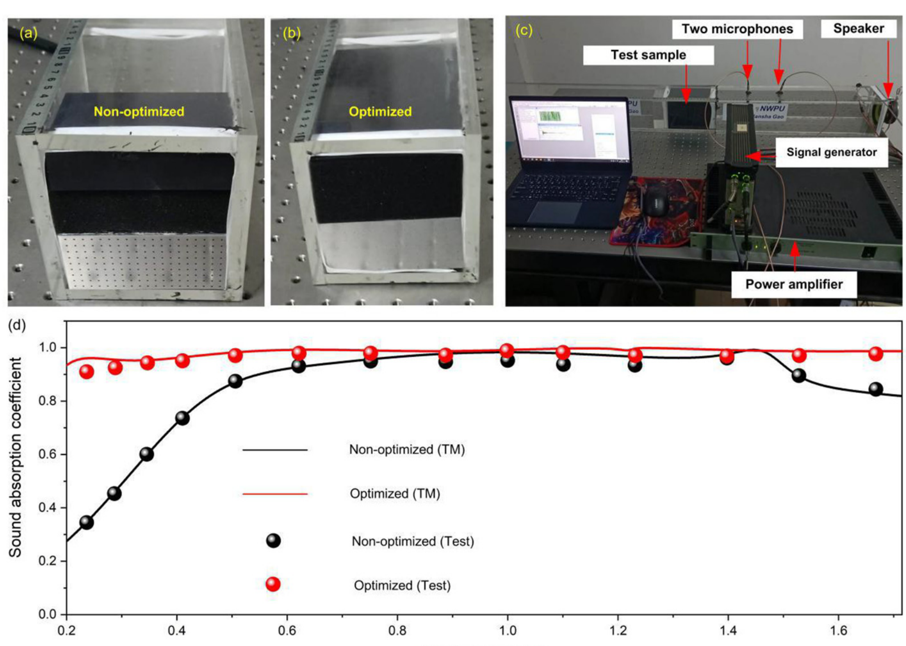

Fig. 5. (a) and (b) Two test samples. (c) Photograph of the acoustic-impedance tube-test equipment. (b) Results of the tube test and the theoretical model compared.

3.3. Verification of the acoustic-impedance with a tube-test

To verify the actual effects of sound-absorption of the UBPSA, two test samples, non-optimized and optimized, were made to perform an acoustic-impedance tube-test, with square cross section – see Fig. 5a and b. To ensure no higher-order waves propagate in the rectangular tube, the effective test frequency of the acoustic impedance tube, with section length 100 mm, did not exceed 1715 Hz. Owing to the physical performance of the loudspeaker, test data above 200 Hz could be confirmed as valid. The rectangular tube, which was made of 10 mm thick rigid acrylic plate, can approximately describe the propagation of plane sound wave in the tube. More specifically, the wall represents a hard sound-field boundary, and the sound/solid coupling should not occur. The sound-absorption test is defined in the ISO 10534–2 standard [34,37]. A random broadband stationary signal from the speaker entered the impedance tube to create a plane wave. The combined transfer function H_{12} for the total sound-field was measured with two microphones, which are referred to as Microphone1 and Microphone2. They were used to measure the sound pressure inside the tube. A distance of 10 mm between two microphones as well as the normal sound-absorption coefficient was calculated using the transfer function method [25]. The instruments, which were used in this test are were: A power amplifier (Brüel & Kjær 2716), a signal generator (Brüel & Kjær 3560B), PULSE software (Brüel & Kjær), microphones (Brüel & Kjær 4187), and a speaker (HiVi-Swans special customization). The above acoustic test system is shown in the Fig. 5c. The test sample was wrapped with waterproof tape to ensure there is no sound-leakage. Five groups of tests were carried out for each sample to minimize random errors. In Fig. 5d, the test data show good agreement with the calculated results of the theoretical model, which validates the sound-absorption effect of the UBPSA in this study.

4. Conclusions

A TLBO algorithm was used to optimize a UBPSA, which could provide 0 to 1750 Hz ultra-broadband sound absorption. The average sound-absorption coefficient for the optimized UBPSA was as high as 0.9255, which indicates that the optimized UBPSA has excellent sound-absorption properties at both medium and low frequencies. The effective density $\rho_{e,i}$ and the bulk modulus $K_{e,i}$ of different effective layers $1^{\#}$-$8^{\#}$ could be adjusted using different lengths of lateral plates. This made it possible to change the overall acoustic impedance on the surface (z_{s11}). The test result for the sound-absorption yielded a value below 1715 Hz. This study confirms that sound-absorption of an UBPSA can be done when a TLBO algorithm is used, which opens new avenues for improved broadband sound-absorbers in the future.

CRediT authorship contribution statement

Nansha Gao: Investigation, Funding acquisition. **Baozhu Wang:** . **Kuan Lu:** Software, Methodology. **Hong Hou:** Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the Aeronautical Science Foundation of China (grant number 20181553015), the National Natural Science Foundation of China (grant number 11704314) and the China Postdoctoral Science Foundation (grant number 2018M631194).

References

[1] Kumar S, Lee HP. Labyrinthine acoustic metastructures enabling broadband sound absorption and ventilation. Appl Phys Lett 2020;116(13):134103.
[2] Cao LY, Yang ZC, Xu YL, Fan SW, Zhu YF, Chen ZL, et al. Flexural wave absorption by lossy gradient elastic metasurface. J Mech Phys Solids 2020;143:104052.
[3] Cao LY, Yang ZC, Xu YL, Fan SW, Zhu YF, Chen ZL, et al. Disordered elastic metasurfaces. Phys Rev Appl 2020;13(1):014054.
[4] Gao NS, Wei ZY, Hou H, Krushynska A. Design and experimental investigation of V-folded beams with acoustic black hole indentations. J Acoust Soc Am 2019;145:EL79-EL83.
[5] Gao NS, Cheng BZ, Hou H, Zhang RH. Mesophase pitch based carbon foams as sound absorbers. Mater Lett 2018;212:243–6.
[6] Ji GS, Fang Y, Zhou J. Porous acoustic metamaterials in an inverted wedge shape. Extreme Mech Lett 2020;36:100648.
[7] Zielinski TG, Venegas R, Perrot C, Cervenka M, Chevillotte F, Attenborough K. Benchmarks for microstructure-based modelling of sound absorbing rigid-frame porous media. J. Sound Vibr. 2020;483:2020.
[8] Tang XN, Yan X. A review on the damping properties of fiber reinforced polymer composites. J Ind Text 2020;49(6):693–721.
[9] Zhao HG, Wang Y, Yu DL, Yang HB, Zhong J, Wu F, et al. A double porosity material for low frequency sound absorption. Compos Struct 2020;239:111978.
[10] Han B, Zhang ZJ, Zhang QC, Zhang Q, Lu TJ, Lu BH. Recent advances in hybrid lattice-cored sandwiches for enhanced multifunctional performance. Extreme Mech Lett 2017;10:58–69.
[11] Opiela KC, Zielinski TG. Microstructural design, manufacturing and dual-scale modelling of an adaptable porous composite sound absorber. Compos Pt. B Eng 2020;187:107833.
[12] Maa DY. Potential of microperforated panel absorber. J Acoust Soc Am 1998;104(5):2861–6.
[13] Yang Z, Dai HM, Chan NH, Ma GC, Sheng P. Acoustic metamaterial panels for sound attenuation in the 50–1000 Hz regime. Appl Phys Lett 2010;96 (4):041906.
[14] Yang Z, Mei J, Yang M, Chan NH, Sheng P. Membrane-type acoustic metamaterial with negative dynamic mass. Phys Rev Lett 2008;101:204301.
[15] Romero-Garcia V, Theocharis G, Richoux O, Merkel A, Tournat V, Pagneux V. Perfect and broadband acoustic absorption by critically coupled sub-wavelength resonators. Sci Rep UK 2016;6:19519.
[16] Long HY, Shao C, Liu C, Cheng Y, Cheng XJ. Broadband near-perfect absorption of low-frequency sound by sub-wavelength metasurface. Appl Phys Lett 2019;115(10):103503.
[17] Donda K, Zhu YF, Fan SW, Cao LY, Li Y, Assouar B. Extreme low-frequency ultrathin acoustic absorbing metasurface. Appl Phys Lett 2019;115:173506.
[18] Guo JW, Zhang X, Fang Y, Jiang ZY. A compact low-frequency sound-absorbing metasurface constructed by resonator with embedded spiral neck. Appl Phys Lett 2020;117:221902.
[19] Ma FY, Chen JY, Wu JH. Time-delayed acoustic sink for extreme sub-wavelength focusing. Mech Syst Signal Pr 2019;141:106492.
[20] Xiong L, Nennig B, Auregan Y, Bi WP. Sound attenuation optimization using metaporous materials tuned on exceptional points. J Acoust Soc Am 2017;142:2288–97.
[21] Park JH, Yang SH, Lee HR, Bin YC, Pak SY, Oh CS, et al. Optimization of low frequency sound absorption by cell size control and multiscale poroacoustics modeling. J Sound Vib 2017;397:17–30.
[22] Chambers AT, Manimala JM, Jones MG. Design and optimization of 3D folded-core acoustic liners for enhanced low-frequency performance. AIAA J 2020;58 (1):206–18.
[23] Wang GG, Tan Y. High performance computing for cyber physical social systems by using evolutionary multi-objective optimization algorithm. IEEE T Cybernetics 2019;49(2):542–55.
[24] Wang YC, Lv JA, Zhu L, Ma YM. Crystal structure prediction via particle-swarm optimization. Phys Rev B 2010;82(9):094116.
[25] Rao RV, Savsani VJ, Vakharia DP. Teaching–learning-based optimization: An optimization method for continuous non-linear large scale problems. Inform Sci 2012;183(1):1–15.
[26] Gao NS, Tang LL, Deng J, Lu K, Hou H, Chen K. Design, fabrication and sound absorption test of composite porous matamaterial with embedding I-plates into porous polyurethane. Appl Acoust 2021;175:107845.
[27] Gao NS, Wu JG, Lu K, Zhong HB. Hybrid composite meta-porous structure for improving and broadening sound absorption. Mech Syst Signal Pr 2021;154:107504.

[28] Gao NS, Luo DD, Cheng BZ, Hou H. Teaching-learning-based optimization of a composite metastructure in the 0–10 kHz broadband sound absorption range. J Acoust Soc Am 2020;148.
[29] Allard JF, Champoux Y. New empirical equations for sound propagation in rigid frame fibrous materials. J Acoust Soc Am 1992;91(6):3346–53.
[30] Li J, Fok L, Yin X, Bartal G, Zhang X. Experimental demonstration of an acoustic magnifying hyperlens. Nat Mater 2009;8(12):931–4.
[31] Yang J, Lee JS, Kim YY. Multiple slow waves in metaporous layers for broadband sound absorption. J Phys D Appl Phys 2017;50(1):015301.
[32] Allard JF, Atalla N. Propagation of sound in porous media: Modelling sound absorbing materials. second ed. John Wiley & Sons Inc; 2009.
[33] Taylor Francis Group. Acoustic absorbers and diffusers: Theory, design and application, third ed. CRC Press Inc., 2017.
[34] Guess AW. Results of impedance tube measurement on the acoustic resistance and reactance. J Sound Vib 1975;40:119–37.
[35] COMSOL Multiphysics® v. 5.5. cn. comsol.com.
[36] Liu CR, Wu JH, Yang ZR, Ma FY. Ultra-broadband acoustic absorption of a thin microperforated panel metamaterial with multi-order resonance. Compos Struct 2020;246:112366.
[37] ISO standard 10534-2-Acoustics-Determination of sound absorption coefficient and impedance in impedance tubes-Part 2: Transfer-function method; 2001.